Repeat Episcleral Plaque Brachytherapy: Clinical Outcomes in Patients Treated for Locally Recurrent Posterior Uveal Melanoma

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• PURPOSE: To report the outcomes of survival, local control, visual acuity, and eye retention in patients treated with repeat episcleral plaque brachytherapy (EPBT) for locally recurrent posterior uveal melanoma (PUM).

• DESIGN: Retrospective, interventional case series.

• METHODS: SETTING: Institutional. PATIENT POPULATION: A total of 1201 patients that underwent iodine-125 (I-125) EPBT as primary treatment for PUM between 1985 and 2015. INCLUSION CRITERIA: Development of locally recurrent disease and retreatment with I-125 EPBT. OBSERVATION PROCEDURES: Clinical records review. MAIN OUTCOME MEASURES: Visual acuity, Kaplan-Meier estimates of survival, local control, metastasis, and loss of the eye over the duration of follow-up.

• RESULTS: Twenty-seven patients (13 men) met our inclusion criteria. Median (range) follow-up from initial treatment was 100 months (14–365 months), while median time to local recurrence was 43 months (9–185 months). Median (range) follow-up after retreatment was 47 months (3–120 months). Kaplan-Meier estimate for local control at 5 years was 77.2% (95% confidence interval [CI], 53.29%–89.91%). All marginal recurrences were successfully retreated whereas 6 of 15 patients with central recurrence developed subsequent re-recurrence following salvage EPBT. Median (range) visual acuity was 20/70 (20/20 to counting fingers at 1 foot) at time of recurrence and declined to counting fingers (20/25 to hand motion) at the most recent follow-up examination. Kaplan-Meier estimate for absence of metastatic disease at 5 years was 78.5% (95% CI, 54.77%–90.70%).

• CONCLUSIONS: Repeat I-125 EPBT offers a viable alternative to enucleation in patients with local recurrence of PUM, yielding high rates of local control with predictable decline in visual acuity. (Am J Ophthalmol 2017;176:40–45. © 2017 Elsevier Inc. All rights reserved.)

POSTERIOR UVEAL MELANOMA (PUM) ARISING IN either the choroid or ciliary body is the most common primary intraocular malignancy of adulthood, with an age-adjusted incidence of 5.1 cases per million.[1] The American Joint Committee on Cancer (AJCC) staging recently unified risk features of PUM prognosis into the tumor, node, and metastasis (TNM) model for anatomic staging.[2] Depending on the tumor's staging classification, patients are treated with either enucleation or one of several globe-sparing treatment modalities: transpupillary thermotherapy (TTT), proton beam radiation therapy (PBRT), stereotactic radiotherapy, or episcleral plaque brachytherapy (EPBT). The specific radioisotope selected for EPBT varies depending on treatment center, with iodine-125 (I-125) being most popular in the United States while ruthenium-106 (Ru-106) is more widely used in Europe and Japan. All globe-sparing treatments carry an inherent risk of local recurrence, ranging from 0% to 55.6% in various case series, depending on which therapeutic modality was used.[3] Increased tumor basal diameter has been consistently identified as the most significant independent risk factor for primary treatment failure.[4–6]

Historically, enucleation has been considered the standard of care for locally recurrent PUMs. At some institutions, including our own, patients with recurrent PUM are given the option of undergoing salvage treatment with EPBT as an alternative to enucleation, provided no contraindication to repeat brachytherapy exists, such as tumor dimensions that exceed maximum plaque size.[7] Owing to the small number of patients who elect this course of treatment, the risks associated with this globe-sparing therapy following primary treatment failure, especially the risk of subsequent local recurrence and distant site metastasis, remain largely unknown. To date only 2 small case series composed of fewer than 15 patients have been published detailing clinical outcomes following salvage EPBT.[8,9] We conducted a retrospective research study of all patients with PUM treated at our institution to investigate if repeat EPBT is a feasible alternative to

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